Tutorial – 11

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1. In 1897, J. J. Thomson "discovered" the electron by measuring the charge-to-mass ratio of "cathode rays" (actually, streams of electrons, with charge q and mass m) as follows:

(a) First he passed the beam through uniform crossed electric and magnetic fields ***E*** and ***B*** (mutually perpendicular, and both of them perpendicular to the beam), and adjusted the electric field until he got zero deflection. What, then, was the speed of the particles (in terms of ***E*** and ***B***)?

(b) Then he turned off the electric field, and measured the radius of curvature, R, of the beam, as deflected by the magnetic field alone. In terms of E, B, and R, what is the charge-to-mass ratio *(q/m)* of the particles?

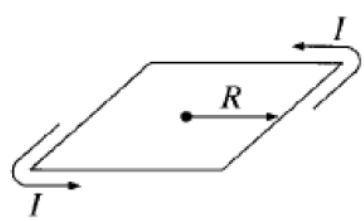
2. A current I flows down a wire of radius *a*.

(a) If it is uniformly distributed over the surface, what is the surface current density *K*?

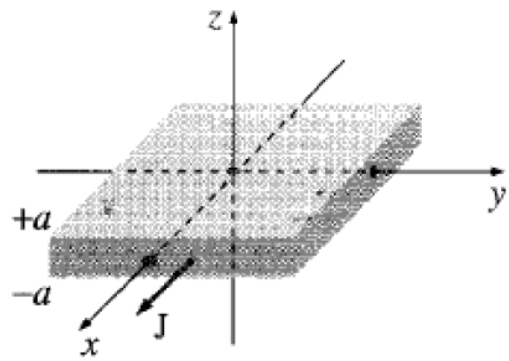
(b) If it is distributed in such a way that the volume current density is inversely proportional to the distance from the axis, what is $J(s)$?

3. Suppose that the magnetic field in some region has the form $\boldsymbol{B} = kz\hat{\boldsymbol{x}}$, where *'k'* is a constant. Find the force on a square loop (side *a*), lying in the y-z plane and centered at the origin, if it carries a current *I*, flowing in a counterclockwise manner.

4. (a) Find the magnetic field at the center of a square loop, which carries a steady current I. Let R be the distance from the center to the side, as shown in the figure.



(b) Find the field at the center of a regular *n-sided* polygon, carrying a steady current *I*. Again let *R* be the distance from the center to any side.

(c) Check that your formula reduces to the field at the center of a circular loop, in the limit $n \to \infty$.

5. A thick slab extending from $z = -a$ to $z = +a$ (and infinite in the x and y directions) carries a uniform volume current $\boldsymbol{J} = J\hat{\boldsymbol{x}}$. Find the magnetic field, as a function of z, both inside and outside the slab.